HireX — Requirement Traceability Matrix (RTM)

Document ID: `HXP-HANDOVER-13`
Platform Version: HireX V0 / Pre-Release Baseline
Target Audience: QA Leads, Product Managers, Engineering Leads
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1. Traceability Matrix

This matrix establishes bidirectional traceability linking **Functional Requirements (FR)**, **Architecture Components & Source Files**, **API Routes**, and **Verified Test Cases**.

Req ID	Requirement Title	Component / Source Files	API Endpoint	Test Case ID	Verification Status
FR-001	Multi-Role Authentication	`src/auth/auth.controller.ts`, `src/candidates/candidate-auth.controller.ts`	`POST /api/auth/sign-in/email`, `POST /api/auth/sign-up/email`	`TC-AUTH-001`, `TC-AUTH-002`	**PASS** (Unit & E2E)
FR-002	Portal Session Isolation	`src/auth/auth.guard.ts`, `src/auth/session-cookie.ts`	All protected routes	`TC-AUTH-004`	**PASS** (E2E Suite)
FR-003	Interviewer Invite Token Onboarding	`src/auth/interviewer-auth.controller.ts`	`GET /auth/interviewer/invite`, `POST /auth/interviewer/invite/accept`	`TC-AUTH-003`	**PASS** (Unit Test)
FR-004	Multivariable Job Search	`src/jobs/jobs.controller.ts`, `hirex-frontend/src/app/jobs/page.tsx`	`GET /jobs`	`TC-CAND-001`	**PASS** (Manual & UI)
FR-005	Dynamic Form Wizard Rendering	`src/applications/applications.controller.ts`, `src/application-form/`	`GET /jobs/:id/application-form`	`TC-CAND-002`	**PASS** (Unit Test)

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FR-006	Application Draft Autosave & Guard	`hirex-frontend/src/app/jobs/[id]/apply/page.tsx`	N/A (Client Local Storage)	`TC-CAND-003`	**PASS** (Verified Code)
FR-007	Guest Candidate Application	`src/applications/applications.controller.ts`	`POST /public/guest-resume`	`TC-CAND-006`	**PASS** (Integration)
FR-008	Recruiter Dynamic Form Builder	`src/jobs/jobs.controller.ts`, `hirex-employer-frontend/src/app/jobs/new/`	`POST /recruiters/jobs`	`TC-REC-001`	**PASS** (Unit Test)
FR-009	AI Resume Relevance Scoring	`src/common/utils/ai-match-scorer.ts`	`POST /applications`	`TC-REC-002`	**PASS** (8 Unit Tests)
FR-010	Strict Forward-Only Pipeline Guard	`src/applications/applications.service.ts`	`PATCH /applications/:id/status`	`TC-SM-001`	**PASS** (E2E Test)
FR-011	Open Interview Pipeline Block	`src/applications/applications.service.ts`	`PATCH /applications/:id/status`	`TC-SM-002`	**PASS** (E2E Test)
FR-012	Weekly Recurring Availability	`src/interviews/availability.controller.ts`	`POST /interviews/availability`	`TC-INT-001`	**PASS** (Unit Test)
FR-013	Self-Service Slot Booking & Concurrency	`src/interviews/booking.service.ts`	`POST /interviews/booking`	`TC-CAND-004`, `TC-CAND-005`	**PASS** (Concurrency E2E)

Req ID	Requirement Title	Component / Source Files	API Endpoint	Test Case ID	Verification Status
FR-014	MeetX Video Room Generation	`src/interviews/providers/schedular-meeting.provider.ts`	Schedular REST API	`TC-MEET-001`, `TC-MEET-002`	**PASS** (Mock & Live)
FR-015	Scorecard & Cultural Fit Submission	`src/interviews/interviews.controller.ts`	`POST /interviews/:id/feedback`	`TC-INT-003`, `TC-INT-004`	**PASS** (Unit & E2E)
FR-016	Branded Offer PDF Generation	`src/offers/offers.service.ts`	`POST /recruiters/offers/generate`	`TC-OFF-001`	**PASS** (Unit Test)
FR-017	Stale Offer Auto-Expiration Cron	`src/offers/offers.service.ts`	`@Cron('0 * * * *')`	`TC-OFF-002`	**PASS** (Unit Test)
FR-018	Multi-File Document Upload	`src/documents/documents.service.ts`	`POST /candidate/documents/upload`	`TC-DOC-001`	**PASS** (Unit Test)
FR-019	Document Rejection & Re-upload Loop	`src/documents/documents.controller.ts`	`PATCH /recruiters/documents/:id/reject`	`TC-DOC-002`	**PASS** (Unit Test)
FR-020	Finalize Hire Compliance Gate	`hirex-employer-frontend/src/app/applicants/[applicationId]/`	`PATCH /applications/:id/status`	`TC-SM-004`	**PASS** (UI & Backend)

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2. Coverage Analysis

- **Total Functional Requirements:** 20 / 20 Mapped (100% Coverage)
- **Total Test Cases Linked:** 80+ Test Cases Across 10 Modules
- **Unverified Requirements:** 0 (All 20 FRs verified via automated test suites or static analysis)